

5	(a)	(i)	Total charge = $nALe$	A1
		(ii)	Current I = total charge / time $= nALe / t$ $= nAve$	M1 A0
	(b)		For each cross-section X and Y, $v = I / nAe$ Since $I_X = I_Y$ (series connection) and $n_X = n_Y$ (same material) $\frac{\text{average drift speed of free electrons at cross-section Y}}{\text{average drift speed of free electrons at cross-section X}} = \frac{A_X}{A_Y}$ $= \frac{\pi d^2 / 4}{\pi (0.69d)^2 / 4}$ $= 2.1$	C1 A1
	(c)	(i)	Resistance = $\frac{\rho L}{A}$ $= \frac{4\rho L}{\pi d^2}$ $= \frac{4(5.1 \times 10^{-7})(50 \times 10^{-2})}{\pi(0.36 \times 10^{-3})^2}$ $R_{AB} = 2.5 \Omega$	M1 A0
		(ii)	1. $L_{CD} = 2L_{AB}$, $d_{CD} = \frac{1}{2} d_{AB}$, $\rho_{CD} = \rho_{AB}$ (same material) Hence, $R_{CD} = 8R_{AB} = 20 \Omega$ $I_{AB} = E / (R_{AB} + R)$ $= 6.0 / (2.5 + 2.5)$ $= 1.2 \text{ A}$ $I_{CD} = E / R_{CD}$ $= 6.0 / 20$ $= 0.3 \text{ A}$ Hence, total current supplied by E = $1.2 + 0.3 = 1.5 \text{ A}$	C1 C1 A1
			2. Potential drop from A to M = $I_{AB}(0.5R_{AB})$ $= 1.2(1.25)$ $= 1.5 \text{ V}$ Potential drop from C to N = $I_{CD}(0.5 R_{CD})$ $= 0.3(10)$ $= 3.0 \text{ V}$ Hence, p.d. between M and N = 1.5 V	M1 A1

c	(a)	239	234	1											
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